

LAB ACTIVITY 3

(Due October 7 @23:59)

Please submit the Jupiter notebook file or .py file with your answers. Please don't remove the error messages from the notebook.

1. Use two variables named "minors" , "adults" and "seniors". Write an if/ else statement that adds 1 to minors if age is less than 18, adds 1 to adults if age is 18 through 64 and adds 1 to seniors if age is 65 or older. Get the age from the user. (2 points)
2. Use the variables *k* and *total* to write a while loop that computes the **sum** of the **squares** of the first 50 counting numbers, and associates that value with *total*. Thus your code should associate $1*1 + 2*2 + 3*3 + \dots + 49*49 + 50*50$ with *total*. Use no variables other than *k* and *total*. (5 points)
3. Given that *n* refers to a positive int use a while loop to compute the **sum** of the **cubes** of the first *n* counting numbers, and associate this value with *total*. Use no variables other than *n*, *k*, and *total*. (5 points)
4. Write a program with for loop that prints all even numbers between two numbers entered by the user. (4 points).
Example: For input range 2 and 10, the output should be 2, 4, 6, 8, 10.
5. Write a program that finds the sum of the digits of a number using a while loop. (3 points). Write another program that finds the sum of the digits of a number using a for loop. (3 points). Example: For input 1234, the output should be $1+2+3+4=10$.